

D853

Meteorology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A meteorologist at a coastal TV station tracks a thunderstorm that grows rapidly as moist air rises and builds towering clouds. The station just upgraded its Doppler radar. What energy source most directly fuels the storm's continued growth?

- A. Latent heat released as water vapor condenses in the rising air.
- B. Heat absorbed during condensation, which cools the cloud and pulls air upward.
- C. Sunlight absorbed directly by the cloud tops.
- D. Heat released as water evaporates from the ground.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A hiking club plans a trip up a 3,000 m peak and asks a forecaster about temperatures. At the trailhead, at 1,000 m elevation, the unsaturated air is 25 °C. The club has 14 members and a certified guide. If a dry parcel rises from the trailhead to 3,000 m without reaching saturation, what is its temperature?

- A. -5 °C
- B. 5 °C
- C. 13 °C
- D. 45 °C

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An unsaturated air parcel cools from 18 °C to 3 °C as it rises. At the dry adiabatic rate, how far did it rise?

- A. 150 km
- B. 2.5 km
- C. 0.3 km
- D. 1.5 km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Two towns at the same latitude, each with about 40,000 residents, sit 800 km apart. Town A is on an ocean coast with frequent onshore winds; Town B is far inland at the same elevation. Town A has a larger airport. How does Town A's climate compare with Town B's?

- A. Town A has a smaller yearly temperature range, with milder summers and winters.
- B. Their climates are the same because they share a latitude.
- C. Town A is always warmer than Town B in every month.
- D. Town A has a larger yearly temperature range.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

On a clear summer afternoon at a busy beach with lifeguard towers every 200 m, the land has heated to 35 °C while the sea surface is 22 °C. A kite festival is scheduled for 3 p.m. Which way does the near-surface wind blow, and why?

- A. From the sea toward the land, because high pressure forms over the hot land.
- B. From the land toward the sea.
- C. No breeze forms, since pressure differences happen only over whole continents.
- D. From the sea toward the land, because warm air rises over the land and cooler sea air flows in to replace it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

At 3 a.m. on a clear night, the land has cooled below the sea-surface temperature. Which breeze forms?

- A. A sea breeze, blowing from sea toward land.
- B. No breeze, since there is no sunlight.
- C. A land breeze, blowing from land toward the sea.
- D. A land breeze, blowing from the sea toward land.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A fishing cooperative on the west coast of South America notices that the usual cold, nutrient-rich upwelling has weakened and the sea surface is 2 °C warmer than normal. The cooperative operates 60 boats. Pacific trade winds have also weakened. What is happening?

- A. A La Niña.
- B. An El Niño: weaker trade winds let warm water spread east and suppress upwelling.
- C. Overfishing by the 60 boats has warmed the water.
- D. A normal seasonal monsoon shift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

During La Niña, what happens in the eastern tropical Pacific?

- A. The sea surface becomes warmer than normal.
- B. Trade winds reverse and blow from west to east.
- C. Trade winds strengthen, upwelling increases, and the sea surface is cooler than normal.
- D. The western Pacific cools, but nothing changes in the east.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A city council member, speaking at a well-attended town hall, says global warming happens because the ozone hole lets more of the Sun's heat in. The town hall was livestreamed on three channels. Which correction is accurate?

- A. Carbon dioxide destroys ozone, and that is what warms Earth.
- B. Greenhouse gases block incoming sunlight so less reaches the surface.
- C. The ozone hole and global warming are exactly the same process.
- D. Warming comes mainly from greenhouse gases absorbing outgoing infrared from Earth's surface; ozone loss mainly lets in more ultraviolet, a separate problem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A research vessel with a crew of 35 measures Arctic sea ice each September. As ice cover shrinks, more dark open ocean is exposed. The ship's hull was reinforced last year. What type of feedback does this create?

- A. None, because albedo matters only over land, never over oceans.
- B. A positive feedback: open water absorbs more sunlight, causing more warming and more melting.
- C. A negative feedback: open water releases heat to space and stops further melting.
- D. A positive feedback, which means the change is good for the climate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which is an example of a negative (stabilizing) climate feedback?

- A. A warmer surface radiates more infrared energy to space, limiting further warming.
- B. Warming thaws permafrost, releasing methane.
- C. Warming increases water vapor, a greenhouse gas.
- D. Snow cover always grows in a warmer world, cooling it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A regional newspaper with a long-running weather column reports that last week set a record low of -28°C in a northern city. A reader with a large following writes that this disproves global warming. Which response is accurate?

- A. Record lows can no longer happen in a warming climate.
- B. Climate is the weather over a single season, so one more season is needed.
- C. A single cold week is weather; climate trends come from long-term averages, usually over 30 years or more.
- D. The reader is right: one record low shows the climate is cooling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A meteorologist ~~at a coastal TV station~~ tracks a thunderstorm that grows rapidly as moist air rises and builds towering clouds. ~~The station just upgraded its Doppler radar.~~ What energy source most directly fuels the storm's continued growth?

Target word: energy source of storm growth **Decides:** Condensation of water vapor in rising air.

A	Correct	Condensation releases heat, keeping the parcel warmer than its surroundings so it keeps rising.
B	Inverted operation	Condensation releases heat; it does not absorb it.
C	Adjacent category	Clouds reflect much sunlight; the storm's engine is latent heat from below.
D	Confusion set	Evaporation absorbs heat; the release happens later, at condensation.

2. Answer B Concept 2

A hiking club plans a trip up a 3,000 m peak and asks a forecaster about temperatures. At the trailhead, at 1,000 m elevation, the unsaturated air is 25 °C. ~~The club has 14 members and a certified guide.~~ If a dry parcel rises from the trailhead to 3,000 m without reaching saturation, what is its temperature?

Target word: parcel temperature at the summit **Decides:** A 2 km rise at the dry rate of 10 °C/km.

A	Wrong stem number	Uses the full 3 km summit height instead of the 2 km rise from the trailhead.
B	Correct	$2 \text{ km} \times 10 \text{ °C/km} = 20 \text{ °C}$ of cooling; $25 - 20 = 5 \text{ °C}$.
C	Confusion set	Uses the moist rate (6 °C/km) for unsaturated air.
D	Inverted operation	Adds the change; rising air expands and cools.

3. Answer D Concept 2 · transfer

An unsaturated air parcel cools from 18 °C to 3 °C as it rises. At the dry adiabatic rate, how far did it rise?

Target word: height from temperature change **Decides:** $15 \text{ °C} \div 10 \text{ °C/km}$.

A	Inverted operation	Multiplies 15×10 instead of dividing.
B	Confusion set	Divides by the moist rate (6 °C/km).
C	Wrong stem number	Uses the ending temperature (3 °C) instead of the 15 °C change.
D	Correct	$18 - 3 = 15 \text{ °C}$; $15 \div 10 = 1.5 \text{ km}$.

4. Answer A Concept 3

Two towns at the same latitude, ~~each with about 40,000 residents~~, sit 800 km apart. Town A is on an ocean coast with frequent onshore winds; Town B is far inland at the same elevation. ~~Town A has a larger airport.~~ How does Town A's climate compare with Town B's?

Target word: coastal vs. inland climate **Decides:** Town A is next to the ocean, with onshore winds.

A	Correct	The ocean's high specific heat slows warming and cooling, damping the coastal range.
B	Dropped element	Drops the ocean's moderating effect.
C	Absolute language	The coast is cooler in summer and warmer in winter, not always warmer.
D	Inverted operation	That describes the continental (inland) town.

5. Answer D Concept 4

On a clear summer afternoon at a ~~busy beach with lifeguard towers every 200 m~~, the land has heated to 35 °C while the sea surface is 22 °C. ~~A kite festival is scheduled for 3 p.m.~~ Which way does the near-surface wind blow, and why?

Target word: direction and cause of the breeze **Decides:** Land is warmer than the sea by day.

A	Confusion set	Right direction, wrong reason: warm rising air creates low pressure over land.
B	Inverted operation	That is the night-time land breeze.
C	Absolute language	Local heating differences routinely drive sea breezes.
D	Correct	Warm land creates lower surface pressure; air moves from higher pressure over the sea.

6. Answer C Concept 4 • transfer

At 3 a.m. on a clear night, the land has cooled below the sea-surface temperature. Which breeze forms?

Target word: night-time breeze **Decides:** Land is now cooler than the sea.

A	Inverted operation	That is the daytime pattern.
B	Dropped element	Drops the temperature difference that still exists at night.
C	Correct	Air rises over the warmer sea; cooler air from the land flows out to replace it.
D	Confusion set	Breezes are named for where they come from; a land breeze blows off the land.

7. Answer B Concept 5

A fishing cooperative on the west coast of South America notices that the usual cold, nutrient-rich upwelling has weakened and the sea surface is 2 °C warmer than normal. ~~The cooperative operates 60 boats.~~ Pacific trade winds have also weakened. What is happening?

Target word: which ENSO phase **Decides:** Weak trades, warm eastern Pacific, weak upwelling.

A	Inverted operation	La Niña brings stronger trades and a cooler eastern Pacific.
B	Correct	All three signs match El Niño.
C	Camouflage	Fleet size does not change sea temperature.
D	Adjacent category	Monsoons are seasonal land-sea circulations, not an interannual Pacific pattern.

8. Answer C Concept 5 • transfer

During La Niña, what happens in the eastern tropical Pacific?

Target word: La Niña conditions **Decides:** The reverse of El Niño.

A	Inverted operation	That is El Niño.
B	Adjacent category	That describes a strong El Niño extreme, not La Niña.
C	Correct	Stronger easterlies push warm water west and pull cold water up in the east.
D	Dropped element	Drops the eastern cooling, which is the defining sign.

9. Answer D Concept 6

A city council member, ~~speaking at a well-attended town hall~~, says global warming happens because the ozone hole lets more of the Sun's heat in. ~~The town hall was livestreamed on three channels~~. Which correction is accurate?

Target word: correct mechanism **Decides:** Greenhouse gases trap outgoing infrared; ozone loss is about UV.

A	Confusion set	Ozone loss is driven by CFCs; CO ₂ warms by absorbing infrared.
B	Inverted operation	They absorb outgoing infrared; they are largely transparent to incoming visible light.
C	Absolute language	They involve different gases, layers, and wavelengths.
D	Correct	The two are distinct processes in different layers with different wavelengths.

10. Answer B Concept 7

A research vessel ~~with a crew of 35~~ measures Arctic sea ice each September. As ice cover shrinks, more dark open ocean is exposed. ~~The ship's hull was reinforced last year~~. What type of feedback does this create?

Target word: feedback type **Decides:** Dark water absorbs more sunlight than ice.

A	Absolute language	Albedo applies to any surface; ice-to-water is a large albedo drop.
B	Correct	Lower albedo amplifies the original warming.
C	Inverted operation	The dominant effect is extra absorption, which amplifies warming.
D	Confusion set	"Positive" means amplifying, not beneficial.

11. Answer A Concept 7 · transfer

Which is an example of a negative (stabilizing) climate feedback?

Target word: identify a damping feedback **Decides:** The effect works against the original warming.

A	Correct	More outgoing radiation offsets the warming that caused it.
B	Confusion set	Looks like a simple consequence, but methane amplifies warming — positive.
C	Inverted operation	More water vapor adds warming — positive feedback.
D	Absolute language	Warming generally reduces snow cover; the claim is false.

12. Answer C Concept 8

A regional newspaper ~~with a long-running weather column~~ reports that last week set a record low of -28°C in a northern city. A reader ~~with a large following~~ writes that this disproves global warming. Which response is accurate?

Target word: weather vs. climate **Decides:** One week is weather.

A	Absolute language	Cold extremes still occur; they become less frequent.
B	Adjacent category	Climate uses decades, not one season.
C	Correct	One event cannot overturn a multi-decade trend.
D	Confusion set	Confuses a weather event with a climate trend.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Latent heat powers storms	When water vapor condenses in rising air it releases latent heat, warming the parcel and driving further uplift in thunderstorms and hurricanes.	Thinking condensation absorbs heat, or that storms run on direct sunlight.	Rapid cloud growth in moist rising air.
2	Adiabatic lapse rates	Unsaturated rising air cools at about 10 °C per km (dry rate); saturated air cools more slowly, about 6 °C per km, because condensation releases heat.	Using the moist rate for unsaturated air, or measuring from sea level instead of the start height.	A starting height and temperature for a rising parcel.
3	Water's high specific heat moderates climate	Oceans warm and cool slowly, so coastal (maritime) places have smaller yearly temperature ranges than inland (continental) places at the same latitude.	Assuming same latitude means same climate.	Coastal vs. inland towns at one latitude.
4	Sea and land breezes	By day, land heats faster, air over land rises, and cooler sea air flows onshore (sea breeze); at night the pattern reverses (land breeze). Breezes are named for where they come from.	Getting the direction backward or naming the breeze for where it goes.	Land vs. sea temperatures at a given time of day.
5	El Niño and La Niña	In El Niño, trade winds weaken, warm water spreads east, and upwelling off South America weakens; La Niña is the reverse, with stronger trades and a cooler eastern Pacific.	Swapping the two phases.	Eastern Pacific water warmer or cooler than normal with a change in trade winds.
6	Greenhouse effect vs. ozone depletion	Greenhouse gases absorb outgoing infrared from Earth's surface and warm the lower atmosphere; the ozone hole is a separate problem that lets more ultraviolet reach the surface.	"The ozone hole lets heat in and causes global warming."	A claim that links ozone and warming.
7	Positive vs. negative climate feedback	A positive feedback amplifies an initial change (melting ice lowers albedo, causing more warming); a negative feedback dampens it. "Positive" means amplifying, not good.	Reading "positive" as beneficial.	A loop in which an effect feeds back on its cause.
8	Weather vs. climate	Weather is short-term conditions; climate is the long-term average and variability, usually over 30 years or more, so a single event cannot confirm or refute a climate trend.	Using one cold week to argue about climate.	A single record event used as evidence about long-term change.

Course scope (public description): Meteorology provides an in-depth exploration of Earth's atmospheric and hydrospheric systems, examining the complex interactions that govern climate and weather patterns. Students will investigate the fundamental concepts of atmospheric circulation, weather systems, and climate dynamics, linking these to the hydrosphere's critical role. The course progresses to a detailed analysis of the Earth's evolving climate, focusing on both natural and anthropogenic factors that have shaped the current state of the atmosphere and hydrosphere. By integrating scientific principles with practical applications, students will gain a comprehensive understanding of how climate systems operate and evolve, preparing them to assess and address future environmental challenges.